



Multiplexing

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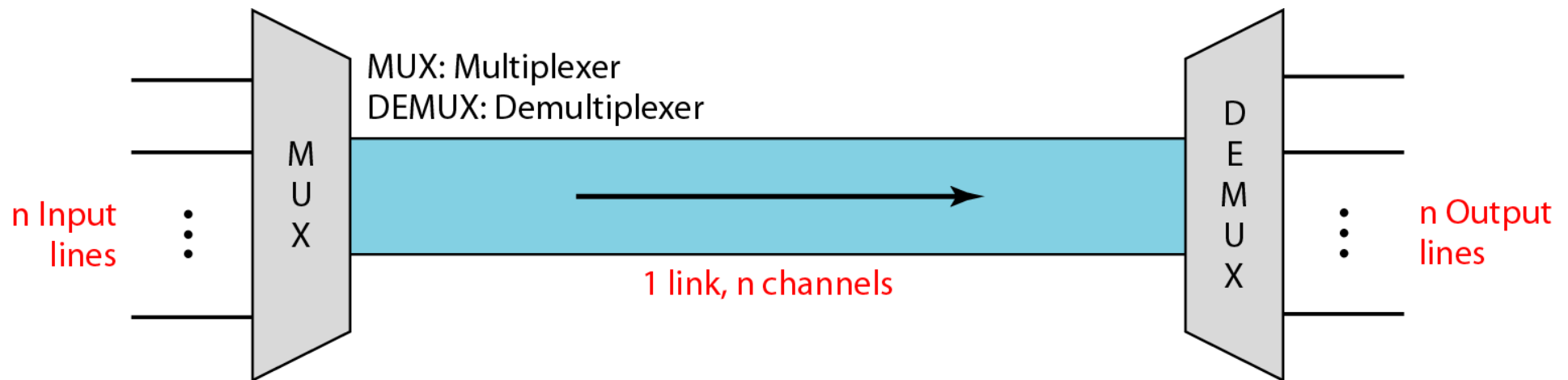


Bandwidth Utilizaion using Multiplexing

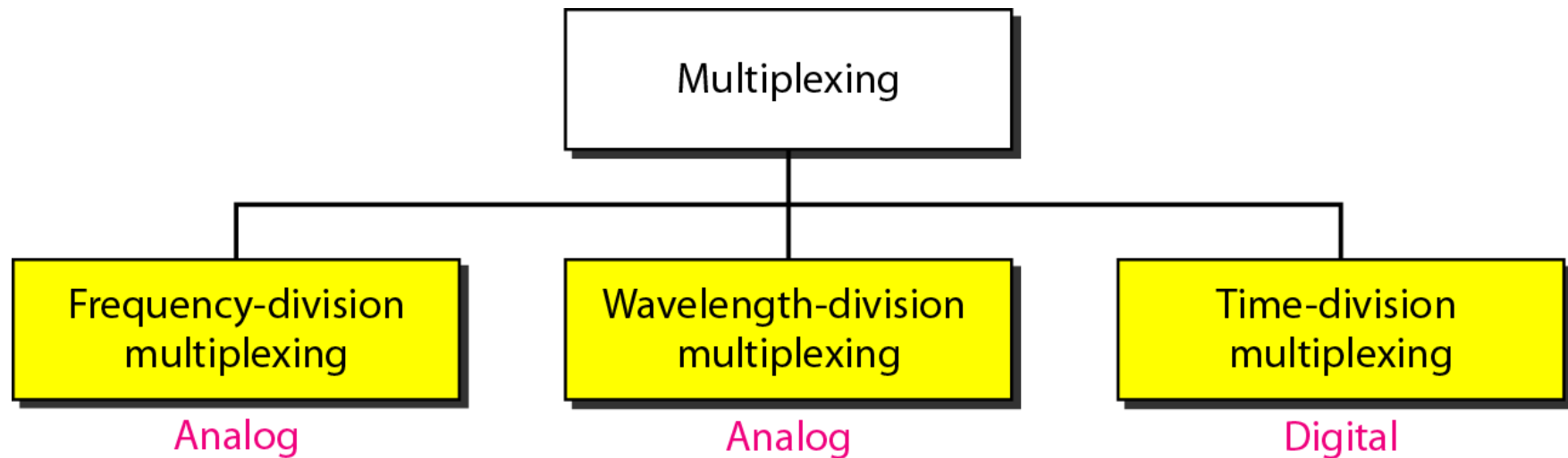
- **Bandwidth utilization** is the wise use of available bandwidth to achieve specific goals
- Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared
- If the bandwidth of a link is greater than the bandwidth needs of the devices connected to it, then the bandwidth is **wasted**
- An efficient system **maximizes** the utilization of all resources

Multiplexing

- Multiplexing is the set of techniques that allows the **simultaneous** transmission of multiple signals across a **single data link**
- Normally, n lines share the bandwidth of one link
- The word channel refers to the portion of a link that carries a transmission between a given pair of lines
- One link can have many (n) channels



Categories of Multiplexing

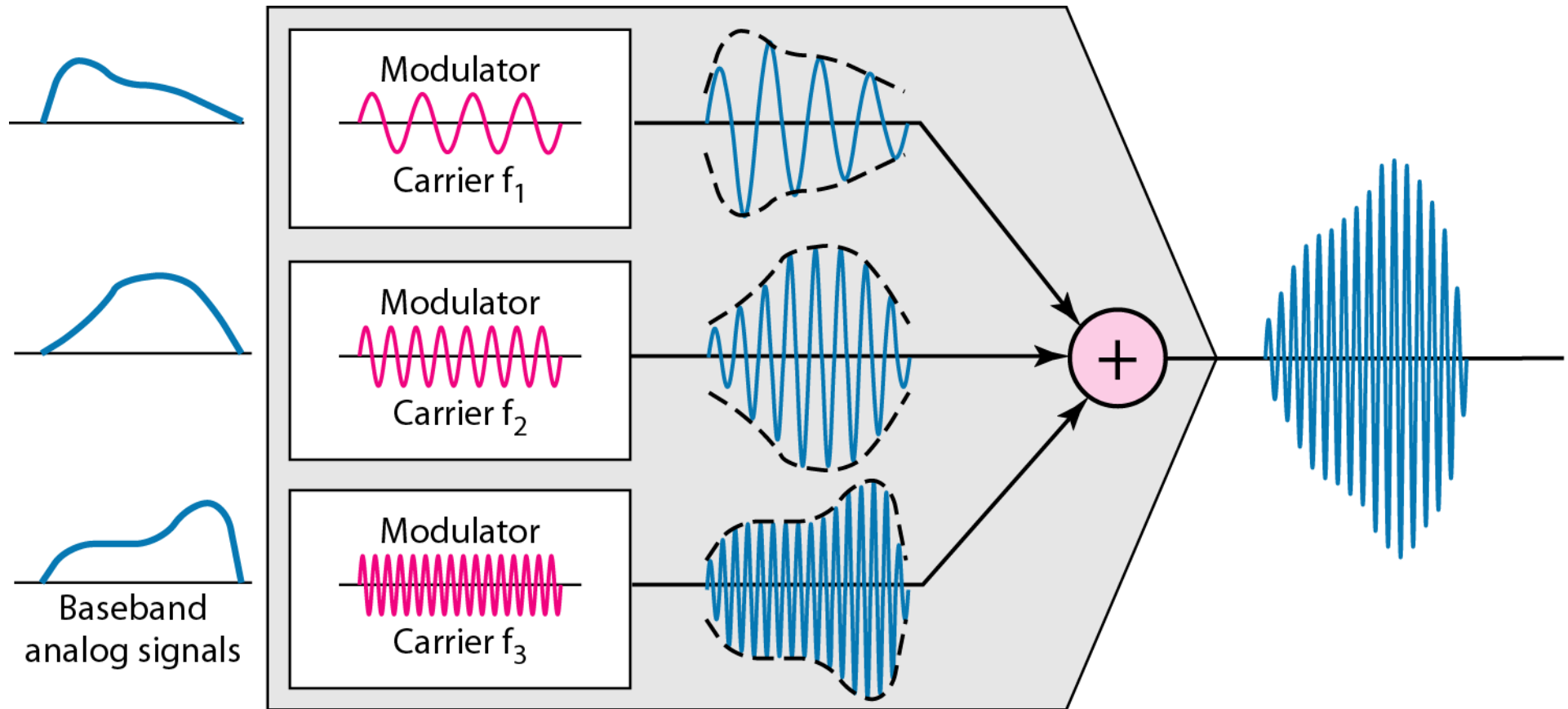


Frequency-Division Multiplexing

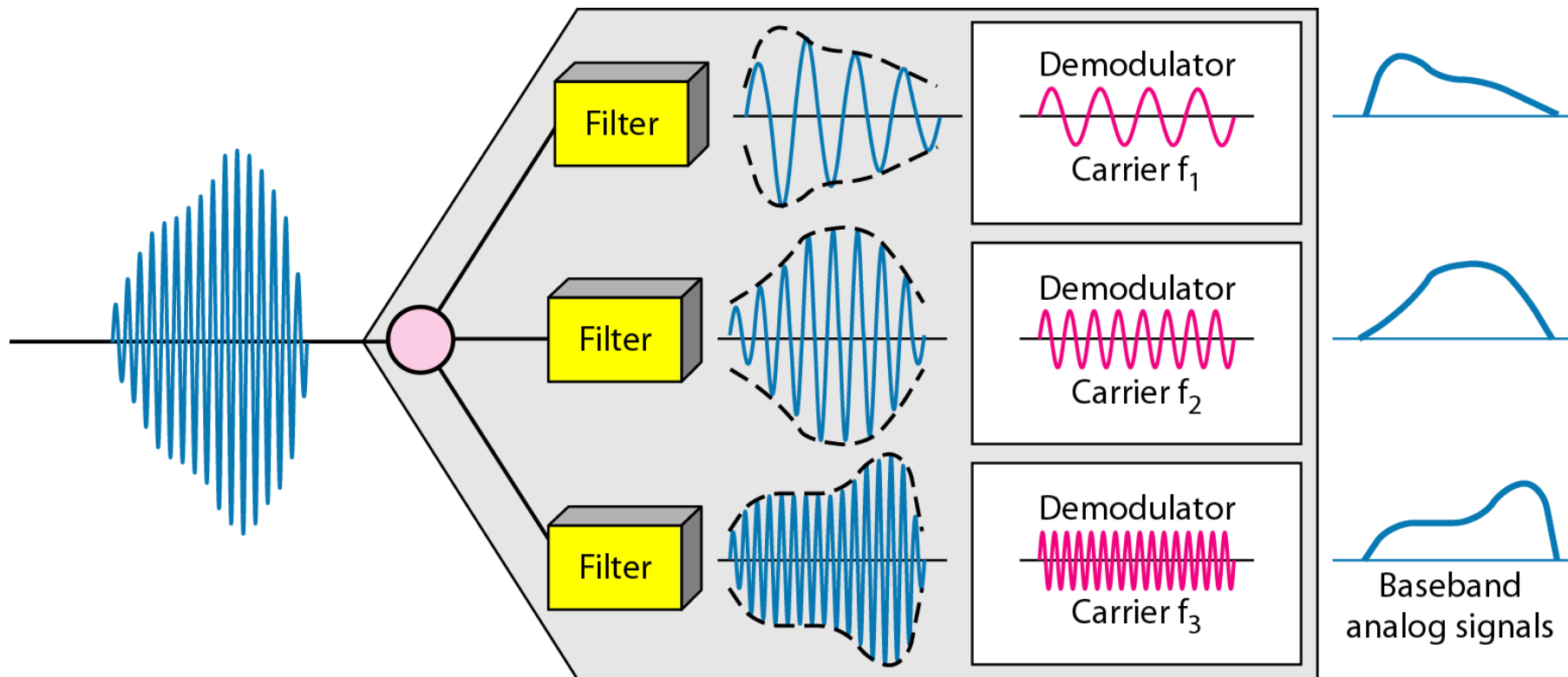
- FDM is an analog technique that can be applied when the bandwidth of a link (in Hertz) is greater than the combined bandwidths of the signals to be transmitted
- In FDM, signals generated by each sending device **modulate** different carrier services
- These modulated signals are then combined into **a single composite signal** that can be transported by the link.



FDM : Multiplexing



FDM : Demultiplexing

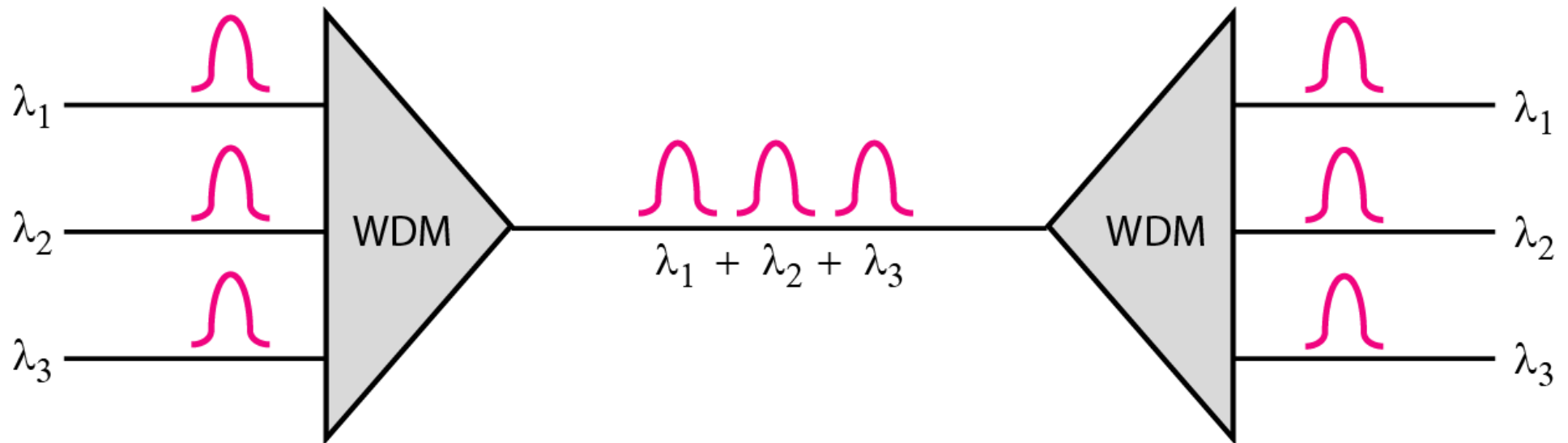




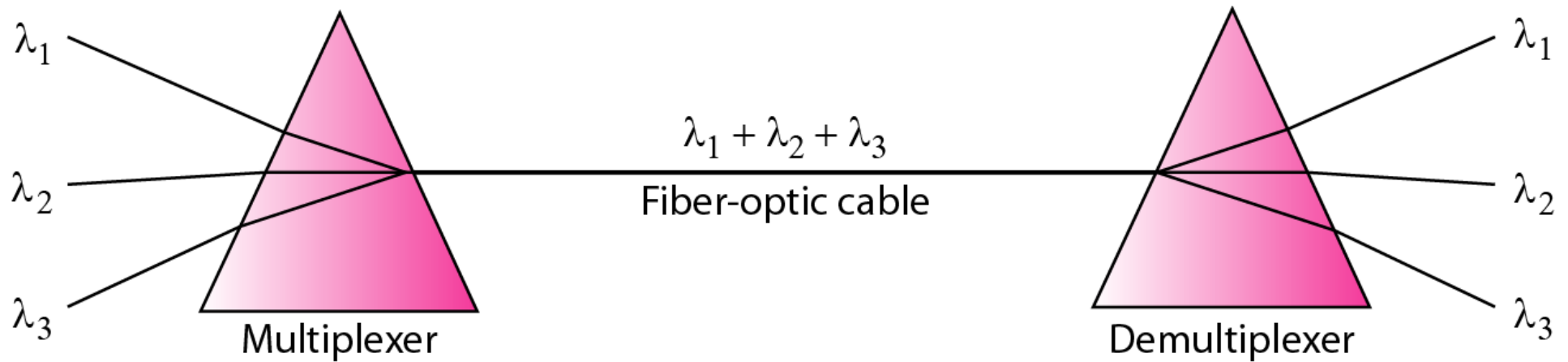
Wavelength-Division Multiplexing

- WDM is designed to use the high-data-rate capability of **fiber-optic cable**
- The optical fibre data rate is higher than the data rate of metallic transmission cable, but using a fibre-optic cable for a single line wastes the available bandwidth
- WDM is conceptually the same as FDM, except that the multiplexing and demultiplexing involve **optical signals** transmitted through fiber-optic channels

Wavelength-Division Multiplexing

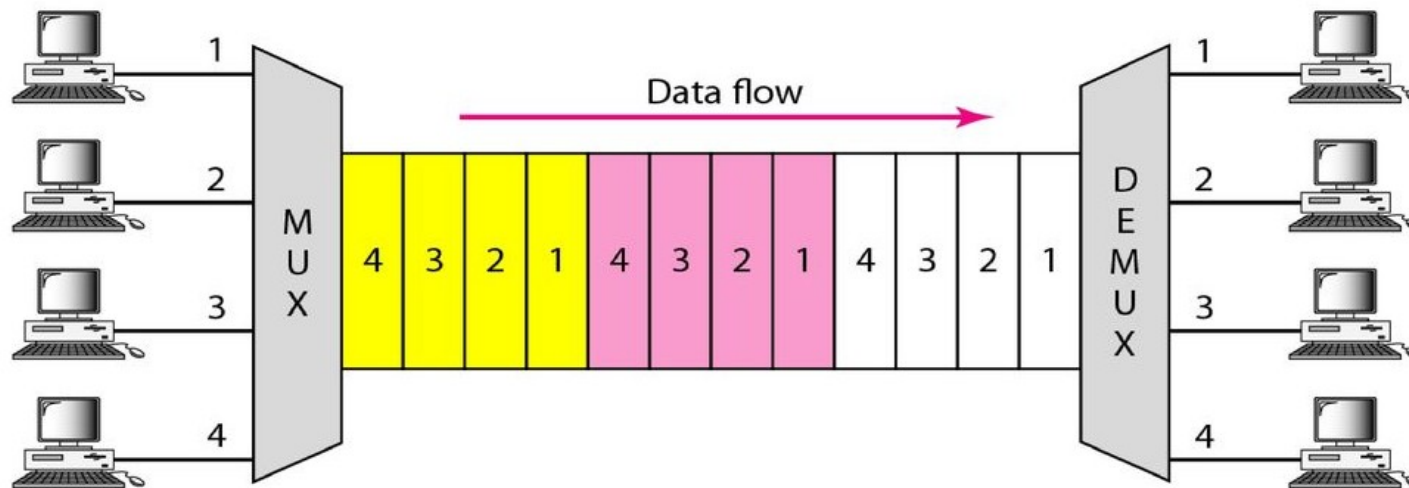


Wavelength-Division Multiplexing



Time-Division Multiplexing

- Time-division multiplexing (TDM) is a **digital process** that allows several connections to share the high bandwidth of a link
- A digital multiplexing technique : digital data from different sources are combined into one **time-shared link**
- Instead of sharing a portion of the bandwidth as in FDM, **time** is shared
- Each connection occupies **a portion of time in the link**



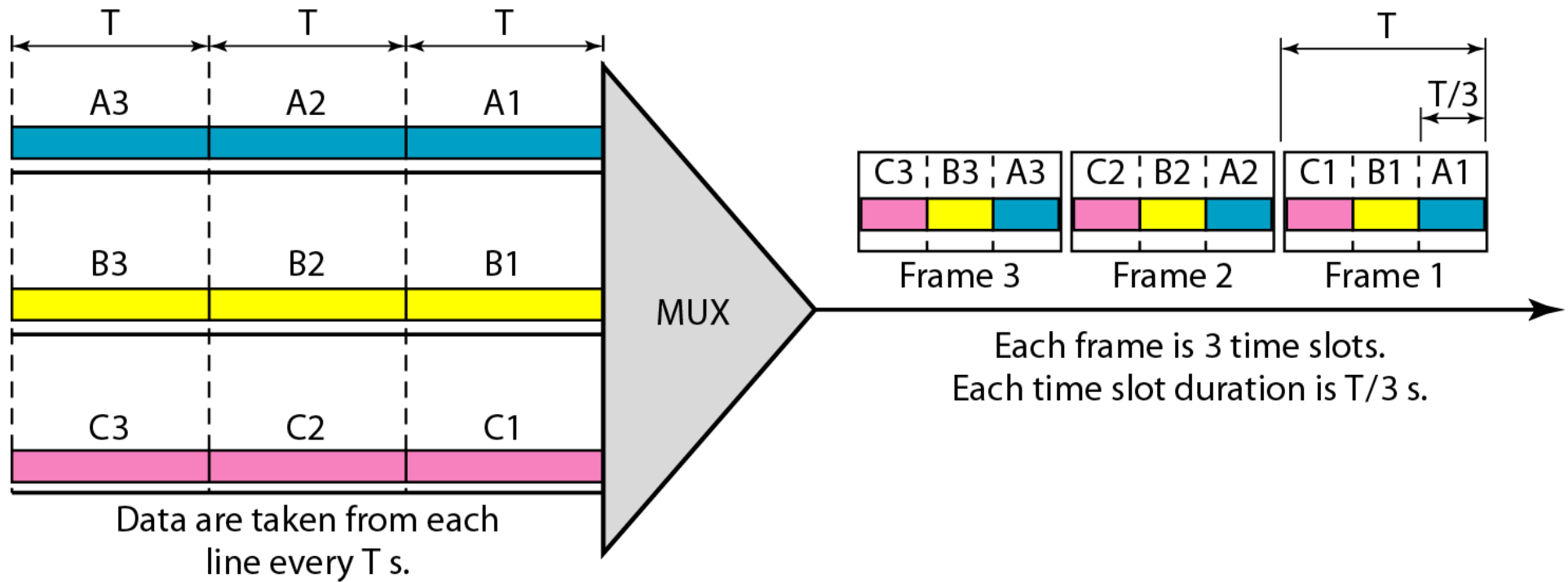


Synchronous TDM

- We can divide TDM into two different schemes
 - **Synchronous**
 - **Statitical.**
- In synchronous TDM, each input connection is divided into units, where each input occupies one input time slot
- A unit can be 1 bit, one character, or one block of data.
- In synchronous TDM, the data rate of the link is n times faster, and the unit duration is n times shorter

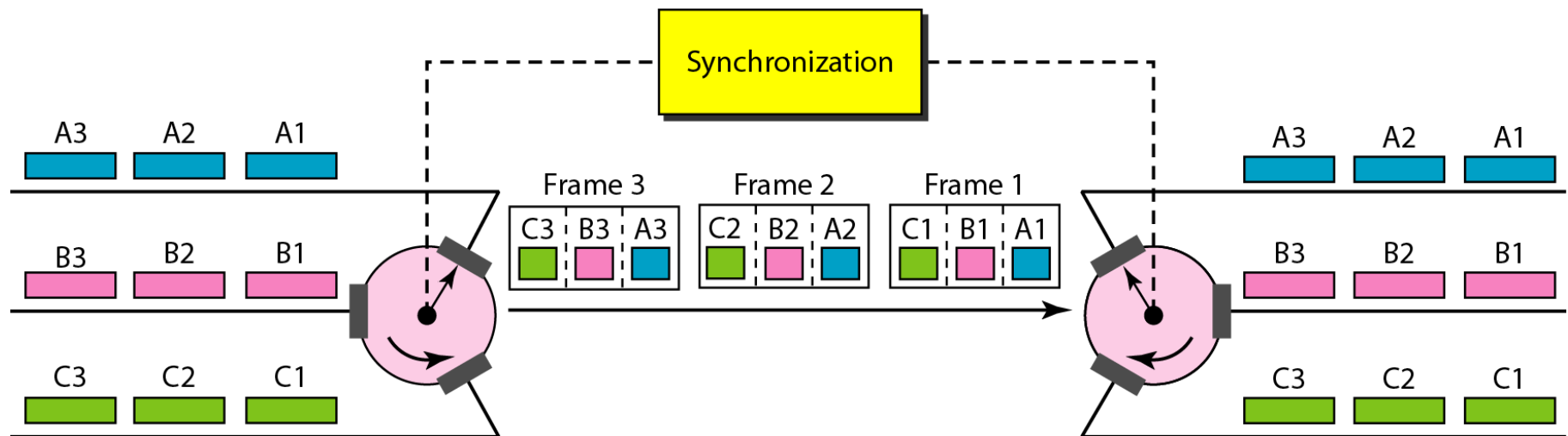


Synchronous TDM



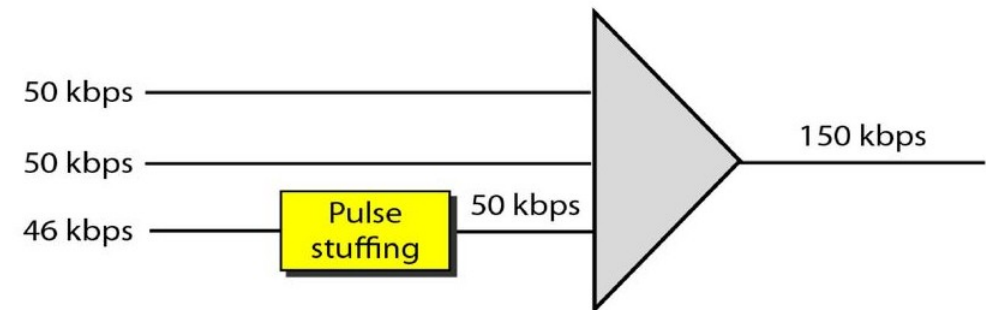
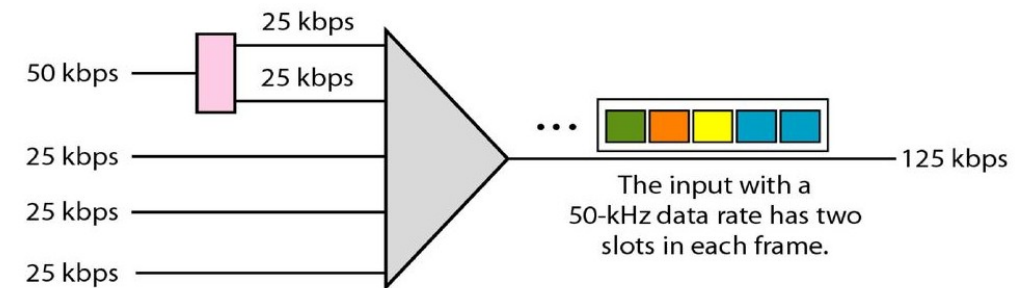
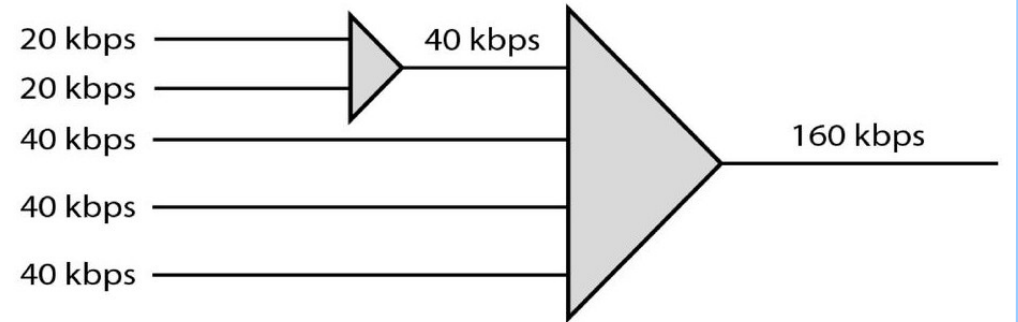
Interleaving

- TDM can be visualized as two fast-rotating switches, one on the multiplexing side and the other on the demultiplexing side.
- The switches are synchronized and rotate at the same speed, but in opposite directions.
- At MUX, as the switch opens in front of a connection, that connection has the opportunity to send a unit onto the path.
- This process is called **interleaving**.
- At DEMUX, as the switch opens in front of a connection, that connection has the opportunity to receive a unit from the path.



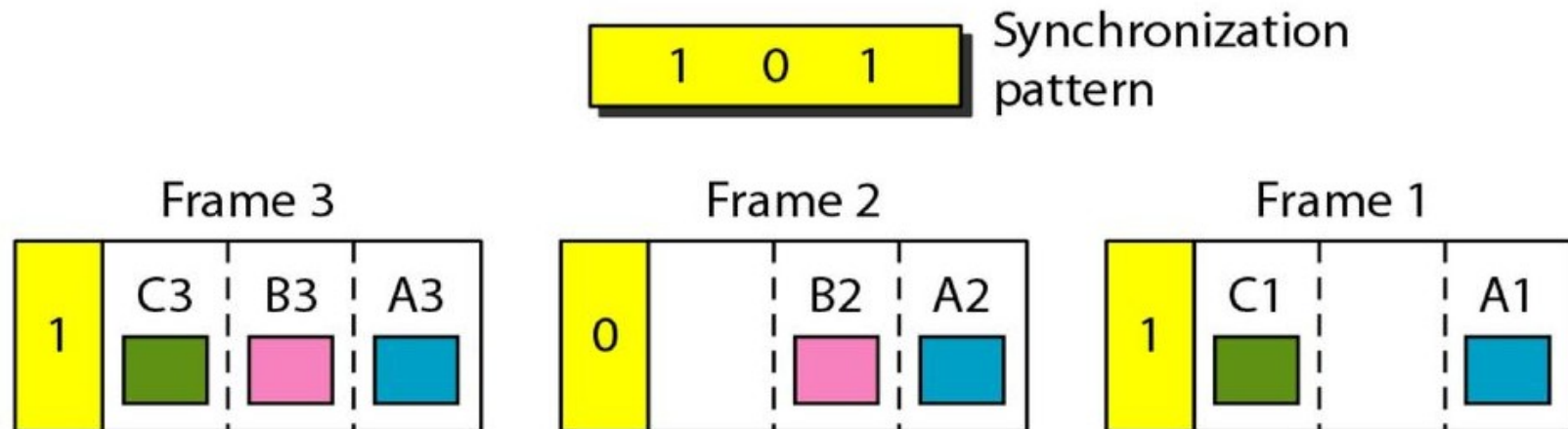
Handling data rate in Synchronous TDM

- if data rates are not the same, three strategies can be used.
- Multilevel Multiplexing
 - It is used when the data rate of an input line is a multiple of others
- Multi-slot Multiplexing
 - Allot more than one slot in a frame to a single input line
- Pulse Stuffing/bit padding/bit stuffing
 - When the bit rates of sources are not multiple integers of each other add dummy bits to the input lines with lower rates



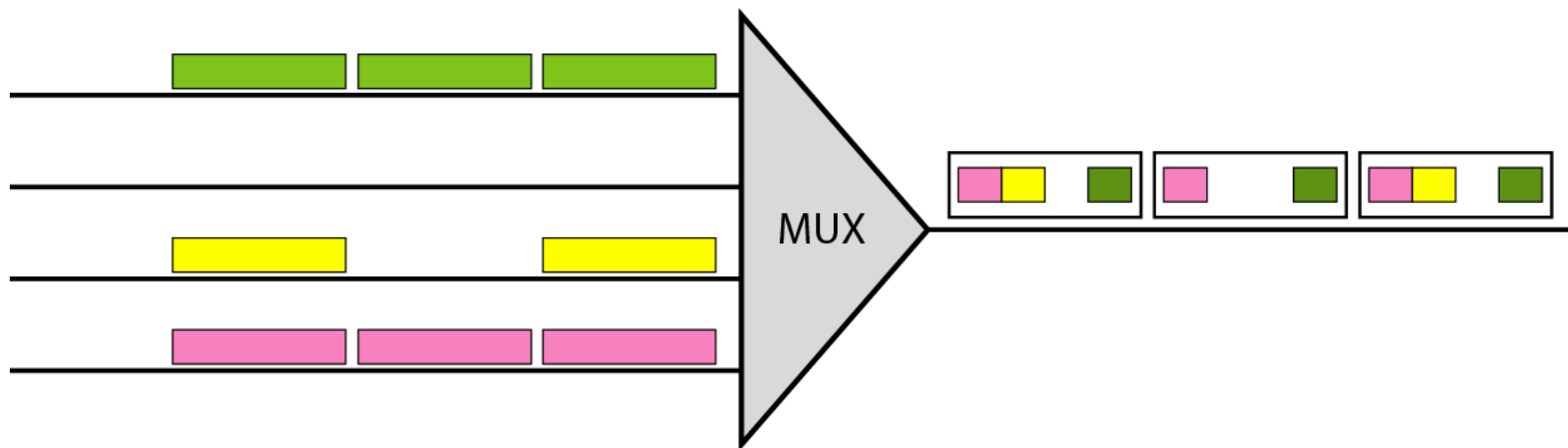
Frame Synchronizing

- If the multiplexer and the demultiplexer are not synchronized, a bit belonging to one channel may be received by the wrong channel
- One or more synchronization bits are usually added to the beginning of each frame (framing bits)
- Allows the demultiplexer to synchronize with the incoming stream so that it can separate the time slots accurately

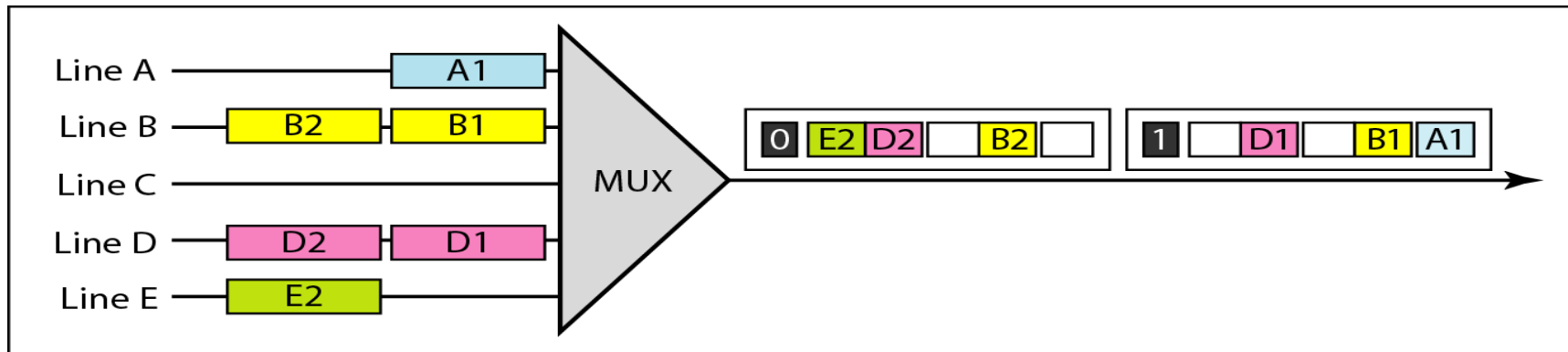


Empty slot problem in Synchronous TDM

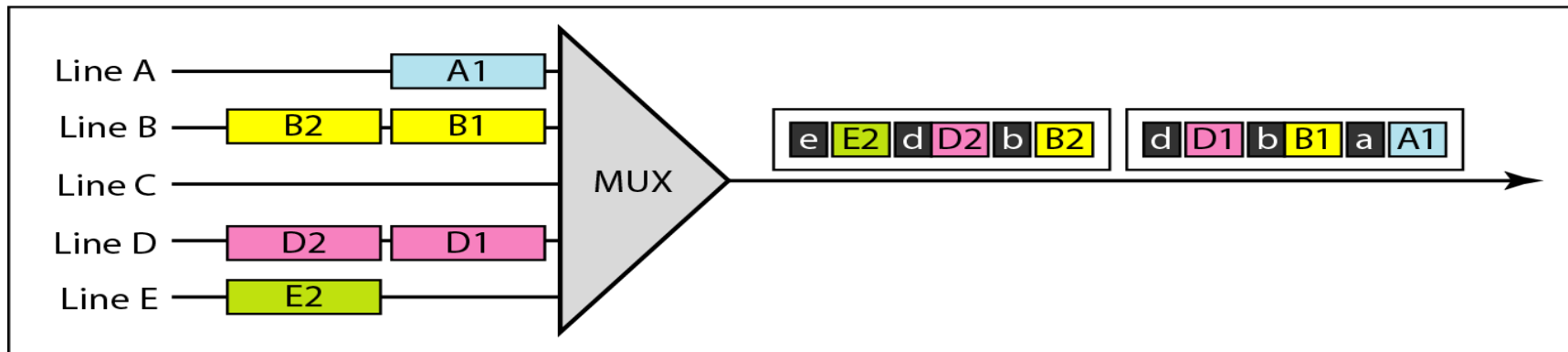
- Synchronous TDM is not as efficient as it could be.
- If a source does not have data to send, the corresponding slot in the output frame is empty.
- **Statistical TDM** can improve the efficiency by removing the empty slots from the frame
- In statistical time-division multiplexing, slots are dynamically allocated to improve bandwidth efficiency.



Synchronous vs Statistical TDM



a. Synchronous TDM



b. Statistical TDM



Statistical TDM

- We need to include the address of the receiver inside each slot to show where it is to be delivered
- The frames in statistical TDM need not be synchronized, so we do not need synchronization bits



Next : Switching